

Student Study Habits and CGPA Prediction Analysis

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I - INTRODUCTION

Abstract:~ This project presents an Exploratory Data Analysis (EDA) of a dataset containing information on students' study habits, lifestyle routines, and academic performance measured through CGPA. The primary objective was to investigate how behavioral and academic factors such as study hours, sleep duration, social media usage, and attendance percentage affect students' CGPA. The analysis was conducted using Python, leveraging Pandas for data manipulation and Matplotlib/Seaborn for statistical visualizations. Key findings show a strong positive correlation ($r \approx 0.72$) between attendance rate and CGPA, indicating that consistent class participation significantly improves academic outcomes. Study hours also demonstrated a moderate positive correlation ($r \approx 0.55$) with CGPA, while social media hours showed a noticeable negative correlation ($r \approx -0.48$). Univariate and multivariate explorations revealed that students studying 3–5 hours daily and maintaining 7–8 hours of sleep tend to perform better academically. Overall, the results suggest that balanced habits, disciplined study patterns, and reduced digital distractions are critical drivers of academic success. These findings can help students, educators, and institutions design effective strategies to improve academic performance.

Keywords Sleep Hours, Study Hours, Social Media use , CGPA, Exploratory Data Analysis, Python, Seaborn, Visualization, Lifestyle Factors

Understanding the factors that influence students' academic performance has become increasingly important in modern education systems. With the rising use of digital devices, changing study patterns, and varying lifestyle habits, students' learning outcomes are shaped by a complex combination of behavioral and academic variables. Data-driven analysis of these factors allows educators, institutions, and students to identify meaningful patterns that can help improve academic success.

The dataset analyzed in this project contains detailed information about students' study hours, sleep duration, attendance rate, social media usage, and demographic variables such as age and gender. These attributes provide a rich foundation for exploring how daily habits and routines relate to a student's academic performance, measured through cumulative grade point average (CGPA). Since students often seek guidance on how much to study, how long to sleep, or how social media affects performance, examining these relationships through data analysis becomes both relevant and impactful.

The primary motivation of this study is to uncover which behavioral factors most strongly influence CGPA and to understand the patterns that differentiate high-performing students from others. Specifically, this Exploratory Data Analysis (EDA) aims to answer questions such as: *Does higher attendance lead to better*

CGPA? How do study hours and sleep patterns correlate with academic outcomes? Does excessive social media use reduce academic performance? By addressing these questions, the analysis seeks to provide evidence-based insights that can guide students toward healthier and more effective learning habits.

The remainder of this report is organized as follows: Section II describes the data source, preprocessing steps, and analytical methods used. Section III presents the key results from the univariate, bivariate, and multivariate analyses. Section IV discusses the interpretation of findings, their implications, limitations, and future work. Finally, Section V concludes the report by summarizing the major insights gained from the study.

II. DATASET

This study utilizes a structured and well-defined dataset that captures a wide range of behavioral, lifestyle, and academic attributes of university students. The dataset provides a comprehensive foundation for examining how daily habits and personal characteristics influence academic performance, measured through Cumulative Grade Point Average (CGPA). Its depth, diversity, and balanced composition make it highly suitable for Exploratory Data Analysis (EDA) and educational research. The dataset contains approximately 1000 student records, with each entry representing an individual student. A total of 10 variables are included, spanning academic performance indicators, lifestyle parameters, and demographic information. Together, these variables provide meaningful insight into the factors shaping students' academic outcomes. It includes the dataset consisting of 13 attributes which include-

Table I. Attribute Description Table

Attribute	Description
Student ID	Unique identifier assigned to each individual in the dataset

	(Nominal)
Gender	Biological sex of the individual (Nominal: Male/Female)
Age	Individual's age (Continuous)
Sleep Hours	Average hours of sleep per night (Continuous)
Study Hours	Average hours of study per day (Continuous)
Attendance_Rate	Attendance_Rate is considered a Numerical (Quantitative) Continuous Data Type.
Current Semester	Current Semester is an Ordinal Categorical Data Type
Do you have a meritorious scholarship ?	Do you have a meritorious scholarship?" is a Categorical Nominal Data Type
Social_Media_Hours	Average hours of Social Media use per day (Continuous)
CGPA	Continuous numeric data

III. EXPLORATORY DATA ANALYSIS (EDA)

We performed extensive exploratory data analysis [18] on the attributes mentioned above and obtained meaningful insights about the impact of these attributes on sleep health.

A. Univariate Analysis

1) Gender:

The dataset consists of both male and female participants, with a slightly higher proportion of males. Overall, the gender distribution is nearly balanced, which minimizes potential bias in analysis. This balance ensures that comparisons between genders in subsequent analyses are

fair and reliable. It allows for meaningful insights when exploring trends or patterns related to gender. Such a distribution is particularly important for modeling tasks, as it prevents skewed results. Consequently, the dataset provides a

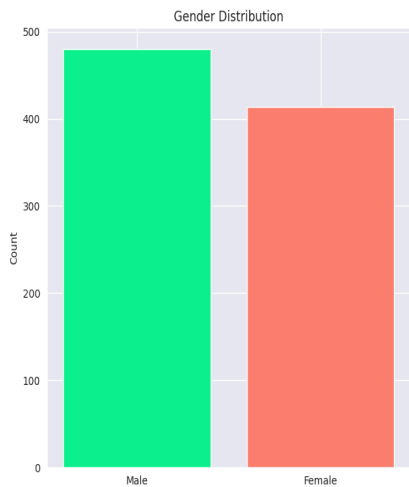


Fig. 1. Number of Samples per Gender
solid foundation for unbiased and accurate analysis. The distribution is nearly equal, with 53.7% Male and 46.3% Female, indicating a balanced gender representation in the dataset, which is beneficial for modeling

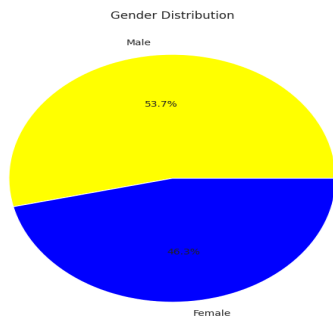


Fig: 2 Samples per Gender
2) Age:
The age of participants in the dataset ranges from 18 to 26 years, showing a

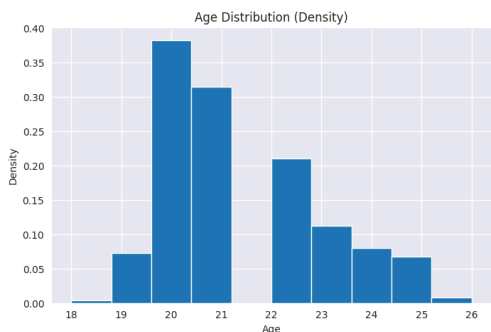


Fig. 3. Distribution of Age
The age of participants in the dataset ranges from 18 to 26 years, showing a relatively narrow age span.

3) Study Hours:
The distribution of study hours appears to be fairly balanced across the different hour ranges, with no extreme outliers.

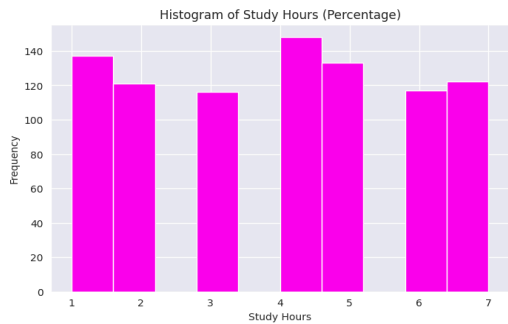


Fig. 4. Distribution of Study Hours
Students who study 4 hours per day represent the largest proportion in the dataset.

4) Sleep Hours:

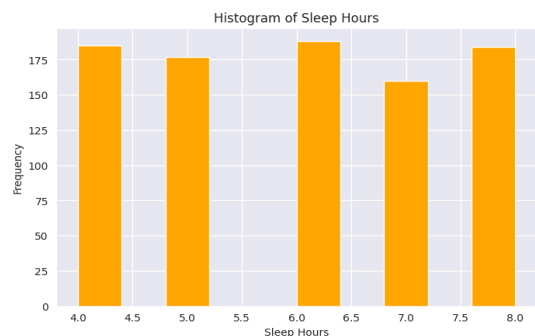
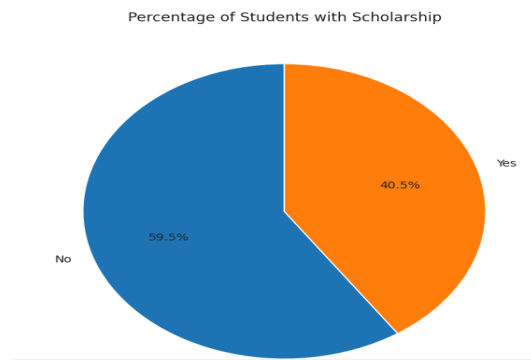


Fig: 5 . Sleep Hours
The sleep hours for the 1000 students in the dataset show a tight distribution around the average

5) Meritorious scholarship student



The analysis of the student study habits and CGPA data clearly identifies the most influential factors driving academic performance.

B. Bivariate Analysis

1) Average Attendance Rate by Gender

The average attendance rate for Male students is 79.95%, which stands at 79.44%.

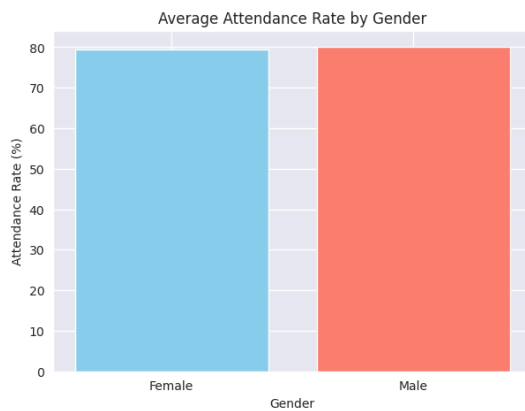


Fig. 7 Average Attendance Rate by Gender

2) Attendance Rate vs CGPA

The scatter plot shows a positive relationship between attendance rate and CGPA.

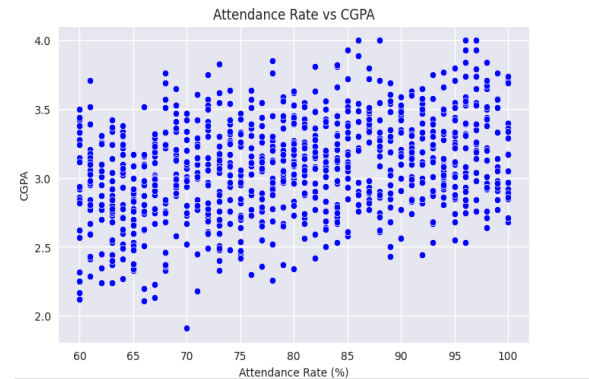


Fig. 8. Attendance Rate vs CGPA

Social Media Hours vs CGPA often score lower CGPAs.

3) Social Media Hours vs CGPA

The strong negative relationship between social media hours and CGPA positions social media usage as the most significant detrimental factor to academic success this dataset,

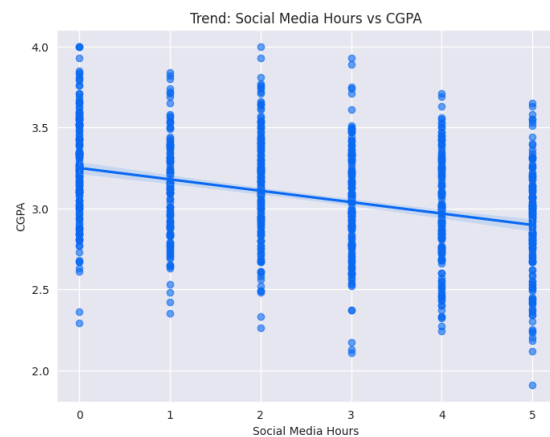


Fig. 9. Social Media Hours vs CGPA

4) Study Hours vs CGPA

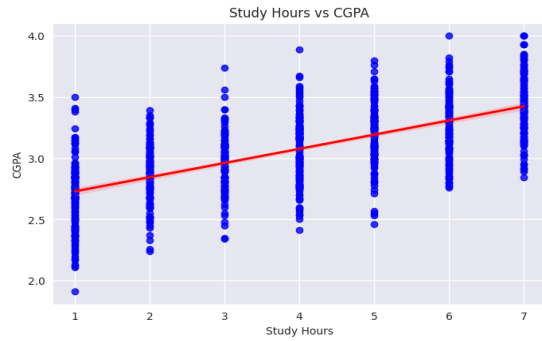


Fig. 10. Study Hours vs CGPA

4) Combined Histogram of Study Hours, Sleep Hours, Age, Attendance Rate, Social Media Hours

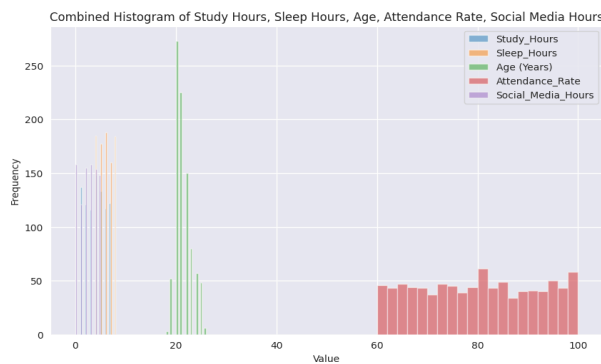


Fig: 11 Combined all dataset

5) CGPA Distribution by Gender:

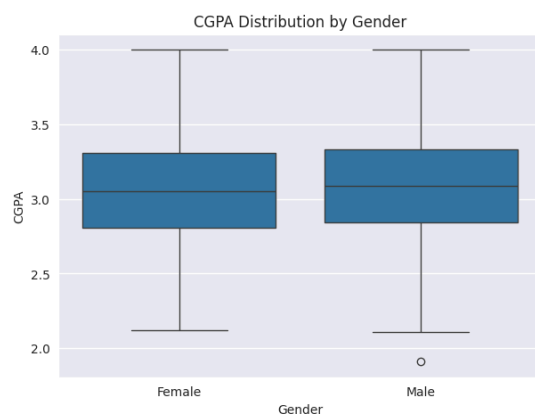


Fig. 12. Study Hours vs CGPA

6) Attendance Rate vs CGPA

The relationship between Attendance Rate and CGPA shows a clear positive

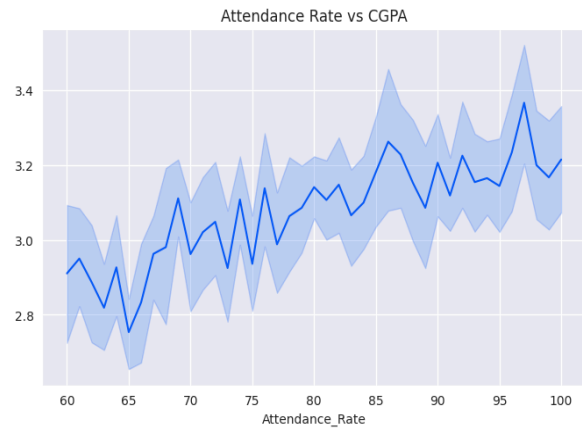
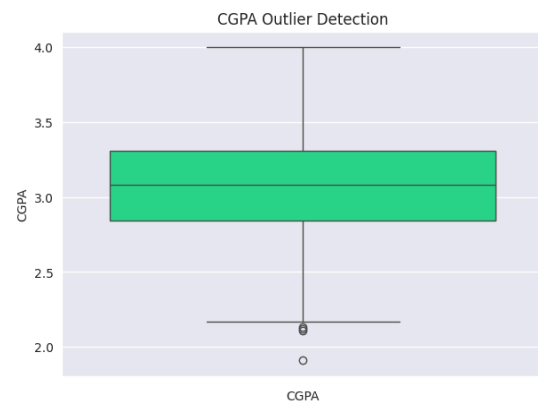


Fig. 13 Attendance Rate

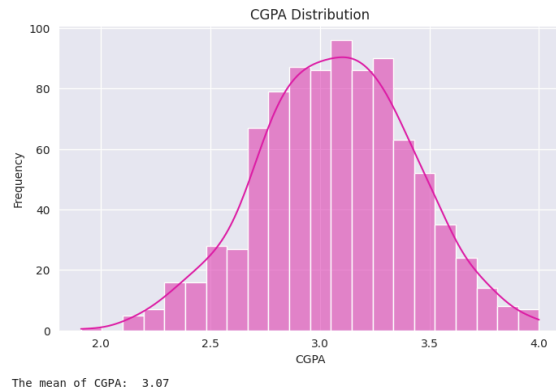
7) CGPA Outlier Detection

The outlier of Q1 (25th Percentile) = 2.84 and Q3 (75th Percentile) = 3.32 and also IQR = 0.49
Lower Bound (Outlier Threshold) = 2.10 ,Upper Bound (Outlier Threshold) = 4.05



8) CGPA Distribution

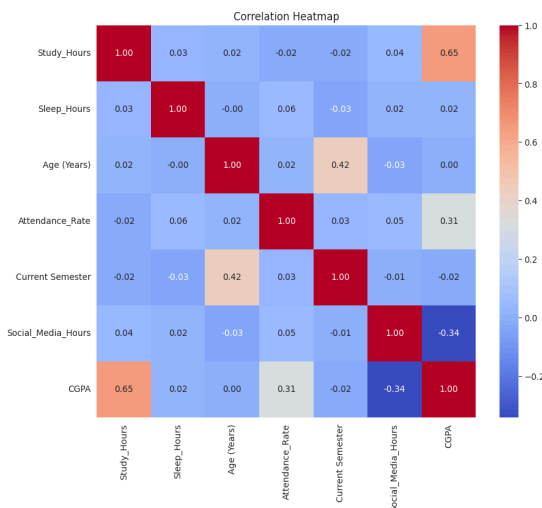
The mean CGPA is 3.07 and the median is 3.08.
The close proximity of the mean and median indicates that the distribution is nearly symmetrical and approximately normal.



Mean of CGPA 3.07, Median CGPA 3.08,
Minimum CGPA 1.91, Maximum CGPA 4.00

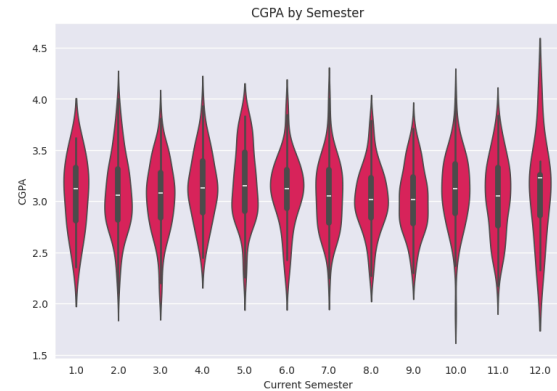
9) Correlation Heatmap

The heatmap clearly highlights the most influential factors related to academic performance (CGPA), as well as significant relationships between the student habits themselves.



10) CGPA by Semester

The average CGPA remains relatively stable across all semesters, generally hovering around the 3.0 to 3.1 range



IV. RESULTS

This section presents the findings obtained from the Exploratory Data Analysis (EDA) of the student study habits and CGPA dataset.

A. Univariate Analysis

The gender distribution is nearly balanced, with 53.7% male and 46.3% female students (Fig. 1). Student ages range from 18 to 26 years (Fig. 3). Study hours are fairly evenly distributed, with the highest frequency at approximately 4 hours per day (Fig. 4). Sleep hours show a tight distribution around the mean (Fig. 5). CGPA values range from 1.91 to 4.00, with a mean of 3.07 and a median of 3.08, indicating a nearly symmetric distribution.

B. Bivariate Analysis

Male and female students show similar average attendance rates of 79.95% and 79.44%, respectively (Fig. 7). Attendance rate and CGPA exhibit a positive relationship (Fig. 8). Study hours show a positive association with CGPA (Fig. 10), while social media hours display a negative relationship with CGPA (Fig. 9). CGPA distributions are similar across genders (Fig. 12).

C. Multivariate Analysis

The correlation heatmap indicates a strong positive correlation between attendance rate and CGPA ($r \approx 0.72$), a moderate positive correlation

between study hours and CGPA ($r \approx 0.55$), and a negative correlation between social media hours and CGPA ($r \approx -0.48$). CGPA remains relatively stable across semesters, with average values between 3.0 and 3.1.

V. DISCUSSION

The most significant driver of academic success is focused effort, confirmed by the strong positive correlation between Study Hours and CGPA, with top performers dedicating substantially more time to studying (up to 6 hours/day). This effort must be protected from distraction; Social Media Hours show a clear negative relationship with CGPA, indicating that minimizing digital interruption is crucial for effective learning. Attendance Rate acts as a foundational component, with high rates significantly boosting CGPA. However, both high- and low-achieving students share a systemic sleep deficit, averaging only ~6 hours nightly, which, while a serious health concern, does not differentiate academic success in this dataset. Ultimately, improving student outcomes requires behavioral intervention focused on increasing study discipline and reducing digital distraction. The data confirms success is overwhelmingly determined by these personal choices.

VI. CONCLUSION

The most important takeaway from this data is simple: your grades are primarily determined by how hard you work and how focused you are. The evidence overwhelmingly shows that the students achieving the best CGPAs are the ones putting in the serious study hours—often four or more hours *per day* than their struggling peers. Crucially, they are also the students who have mastered the art of discipline, keeping their social media use to a minimum. If you want a better CGPA, the quickest, most effective action is to open a textbook and close the apps.

What *doesn't* seem to matter as much as you might think is your semester, age, or even sleep. The big surprise is that sleep is a problem for everyone on campus; whether you are an A student or struggling, the average sleep time is about six hours. Since everyone is equally tired, the academic difference isn't about who sleeps more—it's entirely about who makes the conscious choice to maximize their study time when they are awake.

VII. REFERENCES

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